

BAYESIAN CLASSIFICATION OF DATA FROM GROWTH CURVES

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SUMMARY

A class of Bayesian methods is given for the posterior odds that a set of observations emanated from one of c growth curves. Various assumptions on the parameters are considered. In passing a Bayesian approach to inference on the parameters involved is presented. An example is also included.

1. Introduction

The growth curve model introduced by Potthoff and Roy (1964), is

$$(1.1) E(Y_{p \times N}) = X_{p \times m} \tau_{m \times r} A_{r \times N}$$

where τ is unknown, X and A are known matrices of ranks $m < p$ and $r < N$, respectively. Further, the columns of Y are independent p -dimensional normal variates having a common covariance matrix Σ , i.e., $G(Y|\tau, \Sigma) = N(\cdot; X\tau A, \Sigma \otimes I_N)$ where $\otimes$ denotes the Kronecker product, and $G(\cdot)$ the c.d.f. The reader is referred to Geisser (1970) for further references. In the sequel we will use $g(\cdot)$ and $G(\cdot)$ for p.d.f. and c.d.f., respectively.

In this paper we consider the situation where we have observed sets of data known to have originated from c growth curves, the i -th of which is specified by

$$G(Y_i|\tau_i, \Sigma_i, \Pi_i) = N(\cdot; X\tau_i A_i, \Sigma_i \otimes I_{N_i}), i = 1, 2, \dots, c$$

In addition, a set of observations V , of dimension $p \times K$, which is known to be drawn from one of the c populations $\Pi_1, \Pi_2, \dots, \Pi_c$ with known prior probabilities $q_1, \dots, q_c$, is available. For example, in the data considered by Potthoff and Roy (1964), measurements of girls and boys can be regarded as two growth curves and a further vector observation representing an individual whose sex is unknown is to be classified as either the measurement of a girl or a boy. Here, of course, we assume

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$$(1.2) \quad G(V|\tau_i, \Sigma_i, \Pi_i) = N(\bullet; X\tau_i F_i, \Sigma_i \otimes I_K)$$

where F_i is a known design matrix, usually formed by some columns of A_i , for $i = 1, 2, \dots, c$. For the case where q_i is assumed unknown, arguments similar to Geisser (1964) can be used. It is to be noted that better results should be expected by growth curve classification considered in this paper compared with those of Geisser (1964), if the location structure given by (1.1) is true.

For classification purposes we derive the predictive density of V , $g(V|Y, \Pi_i)$ where Y represents the set of all data $Y_1, \dots, Y_k$, with relevant constant (see Lee (1975)), following the proposal of Geisser (1964, 1966, 1968). In essence the predictive probability that V belongs to Π_i is

$$(1.3) \quad P_i(V \in \Pi_i | Y, q) \propto q_i g(V | Y, \Pi_i)$$

for $i = 1, 2, \dots, c$, $q = (q_1, \dots, q_c)$. We then choose to assign V to Π_i for which (1.3) is a maximum. In this paper we follow Geisser (1970) in the treatment of the prior density of the parameters in assuming a situation where there is little or no prior knowledge regarding the parameters.

Two different covariance structures are considered: (i) arbitrary positive definite (p.d.), and (ii) Rao's simple structure (R.S.S.) $\Sigma = X\Gamma X' + Z\Theta Z'$, where Z is $p \times p - m$ such that $X'Z = 0$, c.f. Lee and Geisser (1972). R.S.S. is a special case of a more general structure considered by Krishnaiah and Lee (1977) wherein, of course, no locational structure was assumed.

In this paper we shall primarily study the problem of classification. But as a spin-off from this work we shall also discuss the estimation of parameters for the aforementioned covariance matrices, since the posterior densities of the parameters will be obtained on the way to deriving the predictive densities of future observations. We shall give some preliminary results in the next section. In Section 3 classification and parametric estimation are considered when Σ is arbitrary p.d. and in Section 4 we discuss the case where $\Sigma = X\Gamma X' + Z\Theta Z'$ obtains. Finally, in Section 5 a numerical example is presented.

2. SOME PRELIMINARIES

Lemma 1.

Let $Y_{p \times N}$, $X_{p \times m}$, $\tau_{m \times r}$, $A_{r \times N}$ be such that the ranks of X and A are $m < p$ and $r < N$, respectively and the rank of Y is $p(\leq N - r)$. Then

$$\begin{aligned} |(Y - X\tau A)(Y - X\tau A)'|^{-N/2} &= \left| \begin{pmatrix} B \\ Z' \end{pmatrix} \right|^{-N} |Z' Y Y' Z|^{-N/2} \\ &\quad \cdot |(X' S^{-1} X)^{-1} + (\tau - \hat{\tau})' G (\tau - \hat{\tau})|^{-N/2} \end{aligned}$$

where

$$B = (X'X)^{-1}X', S = Y(I - A'(AA')^{-1}A)Y',$$

$$t = (X'S^{-1}X)^{-1}X'S^{-1}YA'(AA')^{-1},$$

$$G^{-1} = (AA')^{-1} + T_2'(Z'SZ)^{-1}T_2$$

$$T_2 = Z'YA'(AA')^{-1}$$

and $Z_{p \times p-m}$ is of rank $p-m$ such that $X'Z = 0$.

Proof:

This identity was essentially proved in Sec. 2 of Geisser (1970).

We shall say that B , with dimension $m \times r$, is distributed as $D_{m,r}(\cdot; \Delta, \Lambda, \Sigma, N)$ if it has p.d.f.

$$g(B) = \frac{C_{m,p} \tau^{-rm/2} |\Sigma|^{\nu/2} |\Lambda|^{m/2}}{C_{m,N} |\Sigma + (B - \Delta)\Lambda(B - \Delta)'|^{N/2}}$$

where $\nu = N - r$ and

$$C_{m,p}^{-1} = \pi^{m(m-1)/4} \prod_{j=1}^m \Gamma[\frac{1}{2}(\nu + 1 - j)],$$

Geisser (1966), Lee and Geisser (1972). The general determinantal distribution $D_{m,r}(\cdot)$, also referred to as the matrix t distribution includes the multivariate T distribution, $T(\cdot; \Delta, \Sigma, N)$, as a special case when $r = 1$, $\Lambda = (N-m)^{-1}$. Some properties of $D_{m,r}(\cdot)$ were given in Lee and Geisser (1972).

3. Arbitrary p.d. Σ

Case (3.a); Σ_i unknown, τ_i unknown.

The predictive density of V , with relevant constant, can be obtained from (4.15) of Geisser (1970) as

$$(3.1) \quad g(V|Y_i, \Pi_i) \propto C_i^* |(X'S_i^{-1}X)|^{-\frac{1}{2}(N_i-r)} |G_i|^{m/2} |Z'Y_iY_i'Z|^{-\frac{1}{2}(N_i-m)} \\ \cdot |(X'S_i^{-1}X)^{-1} + B(V - XQ_i)G_i(V - XQ_i)'B'|^{-\frac{1}{2}(N_i+K-r)} \\ \cdot |(Z'(Y_iY_i' + VV')Z)|^{-\frac{1}{2}(N_i+K-m)},$$

for $i = 1, 2, \dots, c$

where

$$\begin{aligned} G_i^{-1} &= M_i^{-1} + (V - \hat{V}_i)' Z (Z' S_i Z)^{-1} Z' (V - \hat{V}_i) \\ (3.2) \quad M_i &= I - F_i' (H_i H_i')^{-1} F_i, \quad H_i = (A_i, F_i), \quad \hat{V}_i = Y_i A_i' (A_i A_i')^{-1} F_i \\ Q_i &= B \hat{V}_i + B S_i Z (Z' S_i Z)^{-1} Z' (V - \hat{V}_i) \end{aligned}$$

$$C_i^* = \prod_{j=1}^m \frac{\Gamma[\frac{1}{2} (N_i + K + 1 - r - j)]}{\Gamma[\frac{1}{2} (N_i + 1 - r - j)]} \times \prod_{j=1}^{p-m} \frac{\Gamma[\frac{1}{2} (N_i + K + 1 - m - j)]}{\Gamma[\frac{1}{2} (N_i + 1 - m - j)]}$$

and S_i is defined as the S in Lemma 1 except that Y is replaced by Y_i and A by A_i . We note that when $K = 1$,

$$C_i^* = \frac{\Gamma[\frac{1}{2} (N_i + 1 - r)] \Gamma[\frac{1}{2} (N_i + 1 - m)]}{\Gamma[\frac{1}{2} (N_i + 1 - r - m)] \Gamma[\frac{1}{2} (N_i + 1 - p)]}$$

The constant of proportionality is mod $\left| \begin{pmatrix} B \\ Z' \end{pmatrix} \right| \pi^{-pK/2}$ and hence irrelevant. The irrelevant constant will be absorbed by the proportionality sign $\propto$ and hence will be omitted from now on.

Estimation with regard to τ_i and Σ_i was considered in Geisser (1970) and Lee and Geisser (1972).

Case (3.b) $\Sigma_i = \Sigma$ but unknown, τ_i unknown.

Using the convenient prior $g(\tau_1, \dots, \tau_c, \Sigma^{-1}) \propto |\Sigma|^{(p+1)/2}$, we have the posterior density of $\tau_1, \dots, \tau_c$ and Σ^{-1}

$$(3.3) \quad g(\tau_1, \dots, \tau_c, \Sigma^{-1} | Y) \propto |\Sigma^{-1}|^{(N-p-1)/2} \cdot \exp \{-\frac{1}{2} \text{tr} \Sigma^{-1} (Y - X\tau A)(Y - X\tau A)'\}$$

where

$$Y = (Y_1, Y_2, \dots, Y_c), \quad \tau = (\tau_1, \tau_2, \dots, \tau_c), \quad N = N_1 + N_2 + \dots + N_c,$$

$$A = \text{Diag}(A_1, A_2, \dots, A_c), \text{ and tr denotes the trace.}$$

Similar to (2.21) of Lee and Geisser (1972) we have a posteriori

$$(3.4) \quad G(\Sigma^{-1} | \tau, Y) = W(\cdot; [(Y - X\tau A)(Y - X\tau A)']^{-1}, N)$$

$$(3.5) \quad G(\tau | \mathbf{Y}) = D_{m,rc}(\cdot; \hat{\tau}, G, (X'S^{-1}X)^{-1}, N)$$

where $\hat{\tau}$, S and G are defined in Lemma 1 and $W(\cdot)$ stands for the Wishart distribution. From (3.4) and (3.5) we obtain,

$$(3.6) \quad E(\Sigma | \mathbf{Y}) = (N - p - 1)^{-1} \{ (\mathbf{Y} - X\hat{\tau}A)(\mathbf{Y} - X\hat{\tau}A)' \\ + (N - m - rc - 1)^{-1} X(X'S^{-1}X)^{-1} X' [\text{tr} G^{-1} A A'] \}$$

We also note that $\hat{\tau}$ is the MLE as well as the posterior expectation of τ , and the MLE of Σ is

$$(3.7) \quad \hat{\Sigma} = N^{-1}(\mathbf{Y} - X\hat{\tau}A)(\mathbf{Y} - X\hat{\tau}A)'.$$

From (1.2), (3.3) it can be shown that the predictive density of V is

$$(3.8) \quad G(V | Y, \Pi_i) = g(V_1 | V_2, Y, \Pi_i) g(V_2 | Y, \Pi_i)$$

where

$$(3.9) \quad G(V_1 | V_2, Y, \Pi_i) = D_{m,K}(\cdot; Q_i^*, G_i^*, (X'S_i^{*-1}X)^{-1}, N + K - rc)$$

$$(3.10) \quad G(V_2 | Y, \Pi_i) = D_{p-m,K}(\cdot; 0, I, Z'Y_i^* Y_i^{*'} Z, N + K - m)$$

Q_i^* , G_i^* are defined in exactly the same way as Q_i , G_i given in (3.2) except Y_i replaced by Y_i^* , A_i by A_i^* , F_i by F_i^* and S_i by $S_i^* = Y_i^* (I - A_i^{*'} (A_i^* A_i^*)^{-1} A_i^*) Y_i^{*'}$, and

$$Y_i^* = (Y_1, Y_2, \dots, Y_{i-1}, Y_{i+1}, \dots, Y_c, Y_i), \tau_i^* = (\tau_1, \tau_2, \dots, \tau_{i-1}, \tau_{i+1}, \dots, \tau_c, \tau_i),$$

$$H_i^* = (A_1^*, F_1^*), A_i^* = \text{Diag}(A_1, A_2, \dots, A_{i-1}, A_{i+1}, \dots, A_c, A_i), F_i^* = (0, 0, \dots, 0, F_i')$$

We note that (3.8) is an extension of (4.15) of Geisser (1970). Hence, similar to (3.1) we have the predictive density of V

$$(3.11) \quad g(V | Y, \Pi_i) \propto | (X'S^{*-1}X) |^{-\frac{1}{2} (N-rc)} | G_i^* |^{m/2} | Z'Y^* Y^{*'} Z |^{\frac{1}{2} (N-m)} \\ \cdot | (X'S^{*-1}X)^{-1} + B(V - XQ_i^*) G_i^* (V - XQ_i^*)' B' |^{-\frac{1}{2} (N+K-rc)} \\ \cdot | Z'(Y^* Y^{*'} + VV') Z |^{-\frac{1}{2} (N+K-m)}$$

Case (3.c) Σ_i unknown, $\tau_i = \hat{\tau}$ unknown.

With the prior $g(\hat{\tau}, \Sigma_1^{-1}, \dots, \Sigma_c^{-1}) \propto \prod_{j=1}^c | \Sigma_j |^{(p+1)/2}$, we have the posterior density

$$(3.12) \quad g(\hat{\tau}, \Sigma_1^{-1}, \dots, \Sigma_c^{-1} | Y) \propto \prod_{j=1}^c |\Sigma_j|^{(N_j - p - 1)/2} \\ \cdot \exp \{ -\frac{1}{2} \text{tr} \bar{\Sigma}_j^{-1} (Y_j - X\hat{\tau}A_j)(Y_j - X\hat{\tau}A_j)' \}$$

By Lemma 1 we have a *posteriori*

$$(3.13) \quad g(\hat{\tau} | Y) \propto \prod_{j=1}^c (X' \bar{S}_j^{-1} X)^{-1} + (\hat{\tau} - \hat{\tau}_j) G_j (\hat{\tau} - \hat{\tau}_j)' \prod_{j=1}^c N_j / 2$$

where $S_j, \hat{\tau}_j, G_j$ are defined in the same way as $S, \hat{\tau}, G$ given in Lemma 1 except Y replaced by Y_j, A by A_j .

We now consider the special case where $c = 2, r = 1$. From (3.12) it is seen that the posterior distribution of Σ_1^{-1} given $\hat{\tau}$ is

$$(3.14) \quad G(\Sigma_1^{-1} | Y, \hat{\tau}) = W(\cdot; [(Y_1 - X\hat{\tau}A_1)(Y_1 - X\hat{\tau}A_1)']^{-1}, N_1)$$

and it can be shown that the posterior density of $\hat{\tau}$ is

$$(3.15) \quad g(\hat{\tau} | Y) = \int_0^\infty P(\hat{\tau} | \Theta, Y) f(\Theta) d\Theta$$

where

$$(3.16) \quad G(\hat{\tau} | \Theta, Y) = T(\cdot; \hat{\tau}^*, \Sigma^*, N), N = N_1 + N_2$$

and for $i \neq j$

$$(3.17) \quad \left\{ \begin{array}{l} f(\Theta) = \tilde{K} \Theta^{2}^{N_1-1} [\alpha^* + (\hat{\tau}_j - \hat{\tau}_i)' \tilde{A}(\hat{\tau}_j - \hat{\tau}_i)]^{-(N-m)/2} |A^*|^{-1/2} \\ \Sigma^* = (N-m)^{-1} \{ \alpha^* + (\hat{\tau}_j - \hat{\tau}_i)' \tilde{A}(\hat{\tau}_j - \hat{\tau}_i) \} A^{*-1} \\ \hat{\tau}^* = \hat{\tau}_i + A^{*-1} X' \bar{S}_j^{-1} X (\hat{\tau}_j - \hat{\tau}_i), \alpha^* = \Theta G_1^{-1} + G_j^{-1} \\ A^* = \Theta X' \bar{S}_i^{-1} X + X' \bar{S}_j^{-1} X, \tilde{A} = \Theta X' \bar{S}_i^{-1} X A^{*-1} X' \bar{S}_j^{-1} X, \end{array} \right.$$

$\tilde{K}$ is a constant such that $\int_0^\infty f(\Theta) d\Theta = 1$. From (3.14), (3.15) and (3.16) we see that

$$(3.18) \quad E(\hat{\tau} | Y) = \int_0^\infty \hat{\tau}^* f(\Theta) d\Theta \\ E(\Sigma_1 | Y) = (N_1 - p - 1)^{-1} \{ (Y_1 - X[E(\hat{\tau} | Y)]A_1)(Y_1 - X[E(\hat{\tau} | Y)]A_1)' \\ + (A_1 A_1') X [\text{cov}(\hat{\tau} | Y)] X' \}$$

Hence, numerically the posterior expectations of $\hat{\tau}$ and Σ_i can be obtained. From (3.15) we obtain a reasonable approximation, see Lee and Geisser (1972), to be

$$(3.19) \quad G(\tau | Y) \approx T(\cdot; \hat{\tau}^*(\hat{\Theta}), \Sigma^*(\hat{\Theta}), N)$$

where $\hat{\Theta}$ maximizes $f(\Theta)$ given in (3.17). We thus have approximately

$$E(\hat{\tau} | Y) \approx \hat{\tau}^*(\hat{\Theta}), E(\Sigma_i | Y) \approx (N_i - p - 1)^{-1} \{ (Y_i - X\hat{\tau}^*(\hat{\Theta}))A_i(Y_i - X\hat{\tau}^*(\hat{\Theta}))A_i' + \frac{N-m}{n-m-2} A_i A_i' X \Sigma^*(\hat{\Theta}) X' \}.$$

We next consider the predictive distribution of V . By combining (1.2) and (3.13) with $c = 2$, $r = 1$ and applying Lemma 1 it can be shown that the predictive density of V is

$$(3.20) \quad g(V | Y, \Pi_i) \propto C_i |Z' Y_j Y_j' Z|^{-N_j/2} |Z'(Y_i Y_i' + VV')Z|^{-(N_i+K)/2} |X'S_j^{-1}X|^{N_j/2} \\ \cdot |X'S_{0i}^{-1}X|^{(N_i+K)/2} \int_0^\infty t^{(N_j/2)-1} [\ell(t)]^{-\frac{1}{2}(N_i+N_j+K)} \\ \cdot |D_i(t)|^{-\frac{1}{2}} dt$$

where

$$(3.21) \quad C_i = \Gamma^{-1} \left(\frac{N_i+K}{2} \right) \Gamma^{-1} \left(\frac{N_j}{2} \right) \prod_{\alpha=1}^p \frac{\Gamma[\frac{1}{2}(N_i+K+1-\alpha)]}{\Gamma[\frac{1}{2}(N_i+1-\alpha)]}$$

$$D_i(t) = tG_j X' S_j^{-1} X + G_{0i} X' S_{0i}^{-1} X$$

$$\ell(t) = 1 + t \{ (\hat{\tau}_j - \hat{\tau}_{0i})' G_j X' S_j^{-1} X [D_i(t)]^{-1} G_{0i} X' S_{0i}^{-1} X (\hat{\tau}_j - \hat{\tau}_{0i}) \}$$

and S_{0i} , $\hat{\tau}_{0i}$, G_{0i} are defined in exactly the same form as S , $\hat{\tau}$ and G given in Lemma 1 except Y replaced by $U_i = (Y_i, V)$, A replaced by $H_i = (A_i, F_i)$. The one-dimensional integral in (3.15) and (3.20) can be executed with any desired accuracy without difficulty through the modern computer, see, e.g., Lee (1975).

Case (3.d) Σ_i known, $\tau_i = \hat{\tau}$ unknown.

The posterior density can be shown to be

$$(3.22) \quad g(\hat{\tau} | \Sigma_1, \dots, \Sigma_c, Y) \propto \prod_{j=1}^c \exp \{ -\frac{1}{2} \text{tr}(X' \Sigma_j^{-1} X) [(\hat{\tau} - \tau_{0j}) A_j A_j' (\hat{\tau} - \tau_{0j})'] \}$$

where τ_{0j} is defined in the same way as $\hat{\tau}$ given in Lemma 1 with S replaced by Σ_j , Y by Y_j and A by A_j . We restrict our consideration to the situation where $c = 2, r = 1$. It can be shown that the posterior distribution of $\hat{\tau}$ is

$$(3.23) \quad G(\tau | \Sigma_1, \Sigma_2, Y) = N(\cdot; \hat{\tau}_0, (\Omega_1 + \Omega_2)^{-1})$$

where

$$\hat{\tau}_0 = (\Omega_1 + \Omega_2)^{-1} (\Omega_1 \tau_{01} + \Omega_2 \tau_{02}), \Omega_i = (A_i A_i') X' \Sigma_i^{-1} X, i = 1, 2$$

The joint density of V and $\hat{\tau}$ is

$$G(V, \hat{\tau} | \Sigma_1, \Sigma_2, Y, \Pi_j) \propto \exp \{ -\frac{1}{2} \text{tr} \Sigma_j^{-1} [(Y_j, V) - X \hat{\tau} H_j] [(Y_j, V) - X \hat{\tau} H_j]' - \frac{1}{2} \text{tr} \Sigma_j^{-1} (Y_j - X \hat{\tau} A_j)(Y_j - X \hat{\tau} A_j)' \},$$

from which it can be shown that the predictive density of V is

$$(3.24) \quad g(V | \Sigma_1, \Sigma_2, Y, \Pi_j) \propto |\Sigma_j|^{-K/2} \exp \{ -\frac{1}{2} \text{tr}(Z' \Sigma_j Z)^{-1} Z' U_j U_j' Z \} \\ \cdot \exp \{ -\frac{1}{2} \text{tr}(Z' \Sigma_j Z)^{-1} Z' Y_j Y_j' Z \} \\ \cdot \exp \{ -\frac{1}{2} \text{tr} X' \Sigma_j^{-1} X [(X' S_{0i}^{-1} X)^{-1} + (\eta_i - \hat{\eta}_{0i}) \tilde{C}_{22 \cdot 1}^{(i)} (\eta_i - \hat{\eta}_{0i})'] \} \\ \cdot \exp \{ -\frac{1}{2} (T_{0i} - T_{0j})' H_i H_i' X' \Sigma_i^{-1} X Q_0^{-1} A_j A_j' X' \Sigma_j^{-1} X (T_{0i} - T_{0j}) \} \\ \cdot |H_i H_i' X' \Sigma_i^{-1} X + A_j A_j' X' \Sigma_j^{-1} X|^{-\frac{1}{2}} \\ \cdot \exp \{ -\frac{1}{2} \text{tr} X' \Sigma_j^{-1} X [(X' S_j^{-1} X)^{-1} + (\eta_j - \hat{\eta}_j) \tilde{C}_{22 \cdot 1}^{(j)} (\eta_j - \hat{\eta}_j)'] \}$$

where S_{0i} is defined as in Case (3.c) and

$$\eta_i = B \Sigma_i Z (Z' \Sigma_i Z)^{-1}, \hat{\eta}_{0i} = B S_{0i} Z (Z' S_{0i} Z)^{-1}, U_i = (Y_i, V)$$

$$Q_0 = H_i H_i' X' \Sigma_i^{-1} X + A_j A_j' X' \Sigma_j^{-1} X, T_{0i} = \hat{\tau}_{0i} - (\eta_i - \hat{\eta}_{0i}) Z' U_i H_i' (H_i H_i')^{-1}$$

$$T_{0j} = \hat{\tau}_j - (\eta_j - \hat{\eta}_j) Z' Y_j A_j' (A_j A_j')^{-1}$$

$$\tilde{C}^{(i)} = \begin{pmatrix} \tilde{C}_{11}^{(i)} & \tilde{C}_{12}^{(i)} \\ \tilde{C}_{21}^{(i)} & \tilde{C}_{22}^{(i)} \end{pmatrix} = \begin{pmatrix} H_i H_i' & H_i U_i' Z \\ Z' U_i H_i & Z' U_i U_i' Z \end{pmatrix}$$

$\tilde{C}^{(j)}$ is defined as $\tilde{C}^{(i)}$ except H_i replaced by A_j and U_i by Y_j and

$$\tilde{C}_{22}^{(i)} = \tilde{C}_{22}^{(i)} - \tilde{C}_{21}^{(i)} \tilde{C}_{11}^{(i)-1} \tilde{C}_{12}^{(i)}.$$

4. Rao's Simple Structure

Case (4.a) $\Sigma_i = X \Gamma_i X' + Z \Theta_i Z'$, Γ_i , Θ_i unknown, $\tau_i = \dot{\tau}$ unknown.

We will consider the case where $c = 2$, $r = 1$. With the convenient prior $g(\Gamma_i^{-1}, \Theta_i^{-1}, \dot{\tau}) \propto |\Gamma_i|^{(m+1)/2} |\Theta_i|^{(p-m+1)/2}$, we have the posterior distribution

$$(4.1) \quad G(\Gamma_i^{-1}, \Theta_i^{-1} | \dot{\tau}, Y) = W(\Gamma_i^{-1}; [(BY_i - \dot{\tau} A_i)(BY_i - \dot{\tau} A_i)']^{-1}, N_i) \\ \cdot W(\Theta_i^{-1}; (DY_i Y_i' D')^{-1}, N_i)$$

and the posterior density of $\dot{\tau}$

$$(4.2) \quad g(\dot{\tau} | Y) = \int_0^\infty g(\dot{\tau} | Y, t) f_1(t) dt$$

where $G(\dot{\tau} | Y, t) = T(\cdot; T_1, \tilde{\Sigma}, N)$, $N = N_1 + N_2$, $T_{1i} = BY_i A_i' (A_i A_i')^{-1}$ and

$$(4.3) \quad \begin{cases} f_1(t) = K t^{\frac{N_1}{2} - 1} [\alpha + (T_{1j} - T_{1i})' \dot{A} (T_{1j} - T_{1i})]^{-(N-m)/2} |\dot{A}^*|^{-\frac{1}{2}} \\ \tilde{\Sigma} = (N-m)^{-1} \{ \alpha + (T_{1j} - T_{1i})' \dot{A} (T_{1j} - T_{1i}) \} \dot{A}^{*-1} \\ T_1 = T_{1i} + \dot{A}^{*-1} (BS_j B')^{-1} (T_{1j} - T_{1i}), \alpha = t(A_i A_i')^{-1} + (A_j A_j')^{-1} \\ \dot{A}^* = t(BS_i B')^{-1} + (BS_j B')^{-1}, \dot{A} = t(BS_i B')^{-1} \dot{A}^{*-1} (BS_j B')^{-1} \end{cases}$$

$\dot{K}$ is a constant such that $\int_0^\infty f_1(t) dt = 1$. It is easily seen that

$$E(\dot{\tau} | Y) = \int_0^\infty T_1(t) f_1(t) dt, E(\Theta_i | Y) = (N_i - p + m - 1)^{-1} D Y_i Y_i' D'$$

$$E(\Gamma_i | Y) = (N_i - m - 1)^{-1} \{ (BY_i - [E(\dot{\tau} | Y)] A_i) (BY_i - [E(\dot{\tau} | Y)] A_i)' + A_i A_i' [\text{cov}(\dot{\tau} | Y)] \}.$$

Hence, numerically the posterior expectations of $\hat{\tau}$ and Γ_i can be obtained. Of course, the posterior expectation of Θ_i is in close form.

A reasonable approximation for the posterior distribution of $\hat{\tau}$ is $G(\hat{\tau}|Y) \approx T(\bullet; T_1(\hat{t}), \hat{\Sigma}(\hat{t}), N)$ where $\hat{t}$ maximizes $f_1(t)$. Hence, approximately $E(\hat{\tau}|Y) \approx T_1(\hat{t})$ and

$$E(\Gamma_i|Y) \approx (N_i - m - 1)^{-1} \{ (BY_i - T_1(\hat{t})A_i)(BY_i - T_1(\hat{t})A_i)' + \frac{N-m}{N-m-2} A_i A_i' \Sigma(t) \},$$

By combining (1.2), (4.1) and (4.2) we have the predictive density

$$(4.4) \quad g(V|Y, \Pi_i) \propto C_i^* |BS_j B'|^{-\frac{1}{2}(N_j-1)} |DY_i Y_i' D'|^{N_i/2} |D(Y_i Y_i' + VV')D'|^{(N_i+K)/2} \\ \cdot |BS_j B'|^{-\frac{1}{2}(N_j-1)} |B_i|^{-\frac{1}{2}(N_i+K-1)} \\ \cdot \int_0^\infty t^{\frac{N_i+K}{2}-1} [h(t)]^{-\frac{1}{2}(N+K)} |tH_i H_i' BS_j B' + A_j A_j' B_i|^{-\frac{1}{2}} dt$$

where

$$C_i^* = \Gamma^{-1} \left(\frac{N_i+K}{2} \right) \Gamma^{-1} \left(\frac{N_i}{2} \right) \prod_{\alpha=1}^m \frac{\Gamma[\frac{1}{2}(N_i+K+1-\alpha)]}{\Gamma[\frac{1}{2}(N_i+1-\alpha)]} \frac{p-m}{\prod_{\alpha=1}^m} \frac{\Gamma[\frac{1}{2}(N_i+K+1-\alpha)]}{\Gamma[\frac{1}{2}(N_i+1-\alpha)]} \quad (4.5)$$

$$B_i = BS_i B' + (BV - T_{1i} F_i)' M_i (BV - T_{1i} F_i).$$

Case (4.b) $\Sigma_i = X \Gamma X' + Z \Theta Z'$, Γ , Θ and τ_i unknown.

By the convenient prior $g(\tau_1, \dots, \tau_c, \Gamma^{-1}, \Theta^{-1}) \propto |\Gamma|^{(m+1)/2} |\Theta|^{(p-m+1)/2}$, we see that the posterior distributions are $G(\tau_i|Y) = D(\bullet; T_{1i}, A_i A_i', BSB', N-r)$,

$$G(\Gamma^{-1}|Y) = W(\bullet; (BSB')^{-1}, N-rc), \quad G(\Theta^{-1}|Y) = W(\bullet; (DYY'D')^{-1}, N), \text{ where}$$

$$S = \sum_{i=1}^c S_i, \quad N = \sum_{i=1}^c N_i, \quad Y = (Y_1, \dots, Y_c).$$

$$\text{Hence, } E(\tau_i|Y) = T_{1i}, \quad E(\Gamma|Y) = (N-rc-m-1)^{-1} BSB', \text{ and}$$

$$E(\Theta|Y) = (N-p+m-1)^{-1} DYY'D.$$

The predictive density of V is easily seen to be

$$(4.6) \quad g(V|Y, \Pi_i) \propto |M_i|^{m/2} |DYY'D' + DVV'D'|^{-\frac{1}{2}(N+K)}$$

$$\cdot |BSB' + (BV - T_{1i}F_i)M_1(BV - T_{1i}F_i)'|^{-\frac{1}{2}(N+K-rc)}$$

For the other cases under this special structure, results are relatively easy to obtain and hence omitted.

5. An Example

For the Potthoff-Roy data, measurements of girls and boys can be regarded as two growth curves. For classification purposes we withhold one vector and use the rest as samples. The design matrices are given as follows:

$$X = \begin{pmatrix} 1 & -3 \\ 1 & -1 \\ 1 & 1 \\ 1 & 3 \end{pmatrix}, A_1 = (1, \dots, 1), A_2 = (1, \dots, 1)$$

where A_1 is 1×10 and A_2 is 1×16 when a girl is being classified. Otherwise A_1 is 1×11 and A_2 is 1×15 .

Using the results given by (4.6) with $q_1 = q_2 = \frac{1}{2}$ we found that out of 27 individuals, 15 are correctly classified and 12 are misclassified. On the other hand, if we use case 8 of Geisser (1964) then out of 27 individuals, 13 are correctly classified. Hence the growth curve classification seems to yield slightly smaller classification errors than the unstructured location setup. Of course, the main purpose here is simply to give a numerical example.

6. Final Remarks

This paper is essentially an extension of Geisser's results (1964) to the growth curves situation. In case locational structure is present, the procedure is expected to yield better classification results.

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